[TAIZEEM NABI]

SOFTWARE ENGINEER • PYTHON & BACKEND SYSTEMS

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PROFESSIONAL SUMMARY

Detail-oriented Backend Software Engineer specializing in **Python, Django, relational database design, and real-time event systems**. Strong foundation in data structures, concurrency handling, and API architecture. Proven ability to architect complex backend solutions including real-time auction bidding engines and high-throughput gaming leaderboard services. Dedicated to writing clean, test-driven, and scalable server-side code.

TECHNICAL SKILLS

Languages:	Python (Proficient), SQL (PostgreSQL, MySQL), Go (Familiar), JavaScript / ES6, Bash
Backend & APIs:	Django, Django REST Framework (DRF), Celery, Redis, WebSockets (Django Channels), RESTful Architecture
Database & Caching:	PostgreSQL, Redis (Sorted Sets, Pub/Sub, Locks), Query Optimization, Transaction Isolation, Indexing
DevOps & Testing:	Docker, Docker Compose, Git & GitHub, Linux/Unix, Pytest, Postman, Locust (Load Testing)
Core Concepts:	Data Structures & Algorithms, OOP, Database Concurrency, Race Condition Prevention, System Design Fundamentals

TECHNICAL PROJECTS

Real-Time Auction & Bidding Engine (eBay Style) — Python, Django REST Framework, Redis, PostgreSQL github.com/taizeem/auction

- Architected an eBay-style proxy bidding engine handling concurrent incoming bids, using **atomic database transactions** (`transaction.atomic()`) and pessimistic locking (`select_for_update()`) to guarantee zero out-of-order bids or race conditions.
- Engineered automated auction closure and sniper-bid extension logic using **Celery Beat and Redis**, dynamically extending auction end times by 60 seconds whenever last-second bids were placed.
- Designed real-time price updates using **Redis Pub/Sub** and WebSockets, reducing client polling overhead and broadcasting live top-bid changes with sub-40ms latency.
- Authored automated test suites with **Pytest** covering edge cases such as concurrent same-amount bids, invalid bid increments, and expired auction rejections with **94% code coverage**.

High-Throughput Real-Time Gaming Leaderboard API — Python, Django, Redis, PostgreSQL, Docker github.com/taizeem/leaderboardAPI

- Engineered a scalable leaderboard service capable of calculating real-time global player ranks across 500K+ player scores using **Redis Sorted Sets (ZSET)** with logarithmic O(log N) lookup complexity.
- Implemented asynchronous write-behind caching to persist score snapshots and player profiles periodically to **PostgreSQL** via background workers, preventing database write exhaustion under intense traffic.
- Built flexible ranking query endpoints with support for global top-N ranks, relative percentile bands, and surrounding-player rankings with p99 response times under **18ms**.
- Stress-tested the API under simulated loads of 2,000+ score-update requests/second using **Locust**, verifying linear scaling and zero dropped updates.

High-Throughput Flash-Sale Ticket Booking Engine — Django 5.1, PostgreSQL 16, Redis 7, Channels, Celery github.com/taizeem/ticket_engine

- Architected a distributed flash-sale ticketing platform engineered to sustain **50,000 req/sec burst traffic**, eliminating lock convoys, DB connection pool starvation, and inventory over-allocation.
- Implemented high-concurrency ticket claims using **PostgreSQL 16** `SELECT FOR UPDATE SKIP LOCKED` within atomic transactions, enabling non-blocking row acquisition across parallel worker processes.
- Protected core database infrastructure using **Redis 7 Lua-scripted Token Buckets** for rate-limiting and a **Redis ZSET virtual waiting room** to buffer incoming user admission queues.
- Dispatched real-time seat availability broadcasts and queue progression via **Django Channels & Daphne (ASGI WebSockets)**, offloading post-booking reconciliation to dedicated **Celery** worker pools.

EDUCATION

[Amar Singh College] — Bachelor of Technology / Science in Computer Science [Srinagar, India] | 2023 – 2026

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Object-Oriented Software Engineering, System Design Basics.

Academic Achievement: Completed Capstone Project on High-Performance Distributed Systems; Dean's List / Academic Merit